PART 1

DATA UNDERSTANDING

1. How many datasets were provided?
   Nine (9) datasets were provided for the ShopSmart Retail data analytics assignment.

- Customers
- Orders
- Products
- Stores
- Returns
- Order_Items
- Suppliers
- Inventory
- Employees

2. The data set provided for the ShopSmart Retail Analytics project represent the following:

- Customers.csv: Contains information about customers who purchase products from ShopSmart Retail Ltd.
- Products.csv: Contains information about the products sold by the company, including product names, categories, brands and prices.
- Orders.csv: Contains records of customer orders and sales transactions.
- Stores.csv: Contains information about the company's retail stores and their locations.
- Returns.csv: Contains records of products or orders returned by customers.
- Employees.csv: Contains information about employees working for the company.
- Suppliers.csv: Contains information about suppliers who provide products to the company.
- Inventory.csv: Contains information about product stock levels and inventory held by the stores.
- Order_Items.csv: Contains details of individual items included in customer orders, such as products, quantities and prices.

3. The Primary Key is the unique ID that identifies each row in that specific table.
    - **Customers:** Customer ID
    - **Products:** Product ID
    - **Orders:** Order ID
    - **Stores:** Store ID
    - **Returns:** Return ID
    - **Employees:** Employee ID
    - **Suppliers:** Supplier ID
    - **Inventory:** Inventory ID

- **Order_Items:** OrderItem ID

4. Foreign Keys are IDs that appear in one table but link back to the Primary Key of another table.

- **Orders Table:** CustomerID (links to Customers), StoreID (links to Stores).
- **Order_Items Table:** OrderID (links to Orders), ProductID (links to Products).
- **Returns Table:** OrderID (links to Orders), ProductID (links to Products).
- **Inventory Table:** ProductID (links to Products), StoreID (links to Stores).
- **Products Table:** SupplierID (links to Suppliers).
- **Employees Table:** StoreID (links to Stores).

5. Relationship between the tables

The data follows a Star/Snowflake Schema where the Orders and Order_Items tables are at the centre:

- Customers → linked to → Orders (One customer can have many orders).
- Stores → linked to → Orders (One store can process many orders).
- Orders → linked to → Order_Items (One order can contain many items).
- Products → linked to → Order_Items (One product can appear in many orders).
- Suppliers → linked to → Products (One supplier provides many products).
- Stores → linked to → Employees (One store employs many staff).
- Stores/Products → linked to → Inventory (Tracks which product is in which store).
- Orders/Products → linked to → Returns (Tracks which product from which order was returned)

PART 4 (XLOOKUP FUNCTION)

You can explain that Supplier Name was the only requested XLOOKUP that could not be completed because the Products table did not contain Supplier_ID, and the Suppliers table did not contain Product_ID or another suitable matching field. Therefore, XLOOKUP could not reliably identify which supplier supplied each product.

PART 8

BUSINESS QUESTIONS

1. Which product generated the highest revenue?

Home Item 400 generated the highest revenue among the products displayed in the Top 10 Products chart, with sales revenue of approximately **₦21,429,952**.

2. Which store performed best?

ShopSmart Lagos was the best-performing store, generating total sales revenue of **₦1,232,192,532**. This indicates that it contributed significantly to ShopSmart Retail's overall

3. Which category sells the most?

The Beauty category generated the highest revenue (**₦1,762,500,500**) according to the Sales by Category PivotTable. This suggests that Beauty products are an important contributor to ShopSmart Retail's sales performance and should be considered when planning promotions and inventory.

4. Which month generated the highest revenue?

December recorded the highest revenue, generating total sales of **₦660,629,527**. This indicates that December was the strongest sales month for ShopSmart Retail during the period analysed. Management could investigate whether festive-season demand, promotions, or increased customer spending contributed to this performance.

5. Which customers spend the most?

Joseph Smith was the highest-spending customer in the Top 10 Customers PivotTable, generating total sales revenue of **₦12,009,782**. This identifies Joseph Smith as a valuable customer, and ShopSmart Retail could consider targeted loyalty rewards and personalised promotions to encourage repeat purchases.

6. Which products are frequently returned?

The category-level return analysis indicates that Home had the highest estimated return rate, at approximately 5.13%, followed by Beauty at 5.08%. This suggests that management should investigate returns in these categories. However, because the Returns dataset does not identify individual products, the analysis cannot confirm which specific products are returned most frequently.

7. Which supplier supplies the highest number of products?

This question could not be answered using the available dataset because the Products table does not contain a Supplier_ID field, and the Suppliers table does not contain a Product_ID or Brand field that links suppliers to products. Therefore, there is no direct relationship between the two tables to determine how many products each supplier supplies. A product-to-supplier mapping or a Supplier_ID column in the Products table would be required to complete this analysis accurately.

8. Which stores have low inventory?

Based on the Inventory PivotTable filtered to show only low-stock items, ShopSmart Abuja (STR017) had the lowest total stock on hand, with 10 units, followed by ShopSmart Kano (STR002) with 11 units, ShopSmart Awka (STR022) with 12 units, ShopSmart Ibadan (STR024) with 13 units, and ShopSmart Lagos (STR013) with 14 units. These stores should be prioritised for inventory review and potential restocking.

9. Which region performs best?

Based on the Sales by Region PivotTable, the South region generated total sales revenue of ₦2,424,534,663. This indicates strong sales performance in the region. Management should examine the factors contributing to this performance and consider applying successful strategies in other regions to improve overall sales.

10. Which product category should management invest in?

Management should consider investing in the Beauty category, which generated the highest revenue of ₦1,762,500,500.

PART 9

SHOPSMART RETAIL: BUSINESS RECOMMENDATIONS REPORT

1. **Key Findings**

1. Beauty is the highest-revenue product category. The Beauty category generated approximately ₦1.76 billion in revenue, making it the strongest-performing category. This indicates a potential area for continued business growth.

2. The South region is the best-performing region. The South generated approximately ₦2.42 billion in revenue, while ShopSmart Lagos was the highest-revenue store, generating approximately ₦1.23 billion. This highlights the importance of understanding and maintaining performance in strong markets.

3. Sales peak in December, and some stores have low inventory. December recorded the highest monthly revenue of approximately ₦660.63 million. The inventory analysis also identified stores with low stock levels, indicating a need for better stock monitoring and replenishment.

2. Business Recommendations

1. Invest strategically in the Beauty category. Management should review demand, profit margins, and stock availability before increasing investment in Beauty products. This will help the business grow revenue while ensuring that additional investment remains profitable.

2. Strengthen inventory management. Management should prioritise stores identified as having low inventory, review their reorder levels, and establish timely replenishment procedures. This can reduce stockouts and help ensure that customers can purchase products when needed.

3. Prepare for seasonal sales peaks. Since December generated the highest monthly revenue, management should use historical sales data to plan stock levels, staffing, and promotional campaigns ahead of the festive season. The business should also investigate the factors contributing to December's strong performance.

3. Risk Identified

Inventory shortages and limitations in the available data present a business risk. Low stock levels may result in lost sales, while the absence of product-level information in the returns dataset limits the ability to identify exactly which products are frequently returned. Management should improve inventory monitoring and strengthen data collection to support more reliable decisions.

4. **Opportunity Identified**

   Growth in the Beauty category and stronger seasonal planning present opportunities for ShopSmart Retail. By understanding customer demand, maintaining adequate stock, and preparing for December's sales peak, management can build on existing revenue performance and identify opportunities for sustainable growth.

5.Conclusion

ShopSmart Retail can improve its performance by investing strategically in high-revenue categories, strengthening inventory controls, and using historical sales patterns to guide planning. Improving data quality will further support evidence-based decisions and long-term business growth.